

Advanced Python Interview Questions with Answers

26. Difference between Python 2 and Python 3?

Python 2 is the older version while Python 3 is the latest version with better Unicode support and print() function.

27. What is a virtual environment in Python?

A virtual environment is an isolated environment used to manage project dependencies separately.

28. What is pip?

pip is Python's package manager used to install libraries.

29. Difference between local and global variables?

Local variables are declared inside functions while global variables are declared outside functions.

30. What are *args and **kwargs?

*args handles multiple positional arguments while **kwargs handles multiple keyword arguments.

31. Difference between is and == ?

is checks memory location while == checks values.

32. What are decorators in Python?

Decorators modify behavior of functions without changing original code.

33. What is multithreading?

Multithreading allows multiple threads to run simultaneously for faster execution.

34. Difference between multithreading and multiprocessing?

Multithreading uses multiple threads with shared memory while multiprocessing uses separate processes.

35. What is file handling in Python?

File handling is used to read and write files.

36. What is the use of with statement?

with automatically closes files after use.

37. What is JSON in Python?

JSON is a lightweight format used for data exchange between applications.

38. How do you handle APIs in Python?

Using libraries like requests, flask, and fastapi.

39. What is Flask in Python?

Flask is a lightweight web framework used to build APIs and web applications.

40. What is Django?

Django is a high-level Python framework used for secure and scalable web applications.

41. Difference between Flask and Django?

Flask is lightweight and flexible while Django is a full framework with built-in features.

42. What is a generator in Python?

Generators produce values one at a time using yield.

43. Difference between return and yield?

return ends the function while yield pauses the function and returns multiple values.

44. What is lambda map filter reduce?

map() transforms data, filter() filters data, and reduce() reduces data to a single value.

45. What is list slicing?

List slicing is used to access part of a list.

46. What is exception hierarchy in Python?

Python exceptions are organized into classes like Exception, ValueError, TypeError, and IndexError.

47. Difference between compile-time and runtime error?

Compile-time errors are syntax errors while runtime errors occur during execution.

48. What is `__name__ == '__main__'` ?

It checks whether a Python file is executed directly or imported.

49. What is an iterator?

An iterator is an object used to iterate through data one element at a time.

50. Why is Python popular in AI?

Python is popular in AI because of easy syntax, AI libraries, community support, and fast development.